Failure-Aware and Confidence-Calibrated Adaptive Retrieval-Augmented Generation System

23CSE399 Project Phase 1 – Approval Presentation

Team ID: 23UG031 **Panel No.:20**

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Agenda

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Screenshot of your guides' approval mail (to go ahead with the presentation)

Problem Statement

Retrieval-Augmented Generation (RAG) improves response quality by retrieving relevant documents from external knowledge sources using embedding similarity. Existing RAG systems primarily rank documents based on similarity scores, assuming higher similarity implies higher relevance.

However, similarity-based ranking alone does not assess the reliability of the retrieval process or detect retrieval failure. As a result, semantically weak, inconsistent, or partially irrelevant documents may be retrieved without warning, leading to hallucinations and degraded response quality.

Despite the critical role of retrieval in downstream generation, current approaches lack a systematic mechanism to quantify retrieval reliability or identify failure signals prior to generation. In particular, there is limited exploration of multi-signal retrieval assessment, such as similarity magnitude, stability of top-k similarity distributions, and agreement among retrieved documents.

This absence of an explicit retrieval failure detection framework represents a significant research gap in RAG systems.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1	H. Soudani, E. Kanoulas, and F. Hasibi	Why Uncertainty Estimation Methods Fall Short in Retrieval-Augmented Generation: An Axiomatic Analysis, 2025	Defines axioms for reliable uncertainty estimation in RAG. Shows token probability and entropy methods ignore retrieval quality and grounding. Emphasizes need for retrieval-aware uncertainty estimation.	Does not propose a practical mechanism to compute retrieval reliability using measurable signals such as similarity score consistency, retrieval variance, or agreement.
2	L. Ngartera, S. Nadarajah, and R. Koina	Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering, 2026	Uses Bayesian embeddings to model retrieval uncertainty and improve reliability.	Does not explicitly detect retrieval failures using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
3	Michael Oehri, Jannis Vamvas, et al.	Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal, 2025	Proposes unified confidence framework using likelihood, entropy, and self-consistency signals. Enables calibrated confidence estimation and risk-controlled refusal in LLM responses.	Does not incorporate retrieval reliability estimation or similarity-based retrieval failure signals in RAG systems.
4	Y. Gao, Y. Xiong, X. Gao, et al.	Retrieval-Augmented Generation for Large Language Models: A Survey, 2023	Provides comprehensive overview of RAG architecture, retrieval methods, and reliability challenges. Identifies uncertainty estimation and retrieval reliability as key limitations.	Does not propose practical methods to quantify retrieval reliability using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
5	S. Es, J. James, L. Espinosa-Anke, and S. Schockaert	RAGAS: Automated Evaluation of Retrieval-Augmented Generation, 2023	Introduces metrics such as context relevance, faithfulness, and grounding to evaluate RAG performance. Enables automated evaluation without human annotation.	Provides evaluation metrics but does not detect retrieval failures or estimate retrieval reliability dynamically during inference.
6	V. Karpukhin et al.	Dense Passage Retrieval for Open-Domain Question Answering, 2020	Introduces dense retrieval using neural encoders and embedding similarity scoring. Improves retrieval accuracy over traditional keyword-based methods.	Uses similarity scores for document ranking but does not analyze similarity patterns to estimate retrieval reliability or detect retrieval failures.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
7	S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling	Corrective Retrieval-Augmented Generation, 2024	Introduces a retrieval evaluator to assess relevance and correctness of retrieved documents. Improves retrieval quality by correcting weak or irrelevant retrieval results.	Requires a separate 'evaluator' model to trigger web searches, adding architectural complexity. It corrects retrieval but doesn't quantify a confidence score for the user.
8	P. Lewis et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020	Introduces RAG framework combining dense retrieval and generation using embedding similarity. Improves factual accuracy by grounding responses in retrieved knowledge.	Does not estimate retrieval reliability or detect weak retrieval before generation.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection, 2023	Introduces reflection tokens to evaluate relevance and usefulness of retrieved documents. Enables the model to critique retrieval quality and improve response grounding.	Relies on 'reflection tokens' and fine-tuning, which increases computational cost and latency. It does not assess the retriever's raw stability before generation begins.
10	O. Khattab and M. Zaharia	ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, 2020	Introduces token-level late interaction similarity scoring using contextual embeddings. Improves retrieval precision and semantic matching between query and documents.	Does not compute retrieval reliability or detect weak retrieval using similarity consistency signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
11	J. Song, X. Wang, J. Zhu, et al.	RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation, 2024	Introduces hallucination-aware tuning pipeline using retrieval-guided training signals. Improves factual grounding and reliability of generated responses in RAG systems.	Does not explicitly estimate retrieval reliability or detect weak retrieval using similarity consistency signals.
12	H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng	Rowen: Adaptive Retrieval-Augmented Generation for Hallucination Mitigation in LLMs, 2025	Introduces adaptive retrieval mechanism to dynamically improve document selection. Enhances retrieval relevance and reduces hallucinations in generated responses.	Does not compute retrieval reliability scores using similarity consistency or retrieval agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
13	Z. Sun, X. Zang, K. Zheng, et al.	ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability, 2025	Uses mechanistic interpretability to analyze internal model activations during generation. Enables detection of hallucinations caused by weak or incorrect grounding.	Focuses on generation analysis but does not estimate retrieval reliability using similarity consistency or agreement signals.

Objectives

The current RAG models employ similarity scores for ranking but lack the ability to identify retrieval failure explicitly.

Similarity score alone does not guarantee retrieval correctness or consistency.

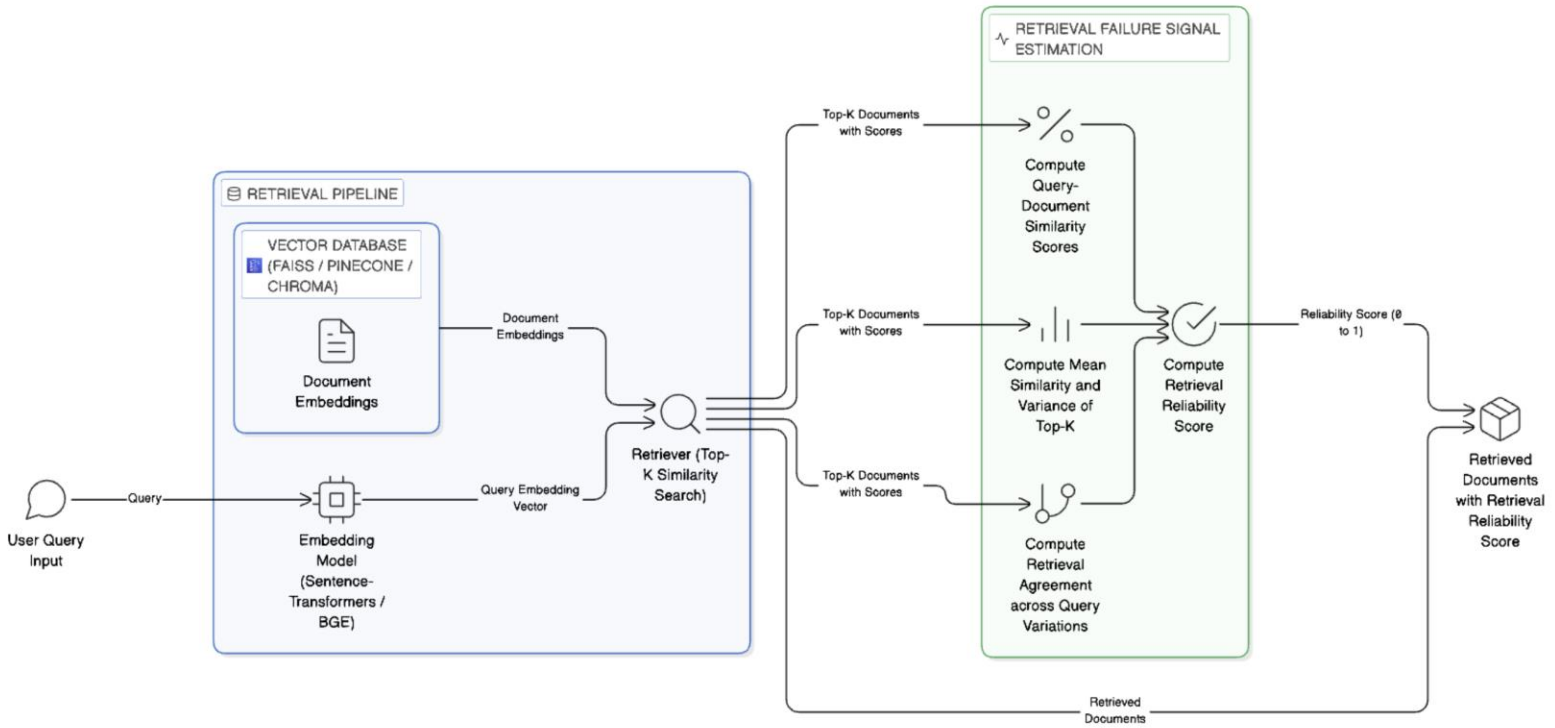
The recent models like Self-RAG and CRAG (Corrective Retrieval-Augmented Generation)assess the quality of retrieval but lack the ability to compute retrieval reliability explicitly from measurable signals.

This results in the potential of poor retrieval going unnoticed, thereby causing unreliable responses.

There is a requirement for a system that computes the retrieval failure signal based on similarity score, top-k consistency, and retrieval agreement.

The plan for the implementation of the estimation of the retrieval failure signal for the identification of retrieval failures is proposed in this work.

Architecture of the Proposed System



Functionalities of the Proposed System

System capabilities:

- Retrieve documents using embedding similarity
- Compute query-document similarity scores
- Analyze top-k similarity consistency
- Compute retrieval agreement signals
- Compute retrieval reliability score using similarity-based failure signals
- Detect weak or unreliable retrieval
- Enable retrieval failure detection

Software/Tools Requirements

Programming Language:
- Python

Frameworks:
- HuggingFace Transformers
- Sentence-Transformers
- LangChain / LlamaIndex
- FAISS / ChromaDB

Models:
- Embedding Model: BGE-large or MiniLM
- Generator Model: Mistral-7B (future integration)

Platform:
- Kaggle GPU Environment

Data Set

Source of data set:

URL:

https://nlp.cs.washington.edu/triviaqa/

Description:

TriviaQA is a reading comprehension dataset containing over 650K question-answer-evidence triples. TriviaQA includes 95K question-answer pairs authored by trivia enthusiasts and independently gathered evidence documents, 6 per question on average, that provide high quality distant supervision for answering the questions.

Timeline

- Abstract Submission • 5/2/2026
- Literature Survey • 17/2/2026
- System Architecture Design Finalization • 5/3/2026
- Embedding Model Integration and Vector Database Setup • 20/3/2026
- Basic Retrieval Pipeline Implementation • 5/4/2026
- Implementation of Retrieval Similarity Score and Top-K Similarity Consistency Signal • 15/4/2026
- Implementation of Retrieval Agreement Signal • 30/4/2026

Conclusion

This work proposes a failure-aware RAG framework to improve system reliability.

This phase focuses on detecting retrieval failures using similarity score, top-k consistency, and retrieval agreement signals.

The system estimates retrieval reliability and enables early detection of weak retrieval.

This forms the foundation for future confidence estimation and adaptive RAG systems.

References

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[12] H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng, “Rowen: Adaptive Retrieval-AugmentedGeneration for Hallucination Mitigation in Large Language Models,” arXiv preprint, 2025.

THANK YOU